Score: $\quad\quad\quad\quad$ **/ 10**

Name (Print): ______________________

Calculus I (MATH 2200)
Spring 2020
Wednesday, 02/26/2020
Quiz 5: Section 3.3, 3.4

Name (Print): ______________________

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Section 3.3

1. Find the indicated derivatives:

$\boxed{2 \text{ pts}}$ (a) $\dfrac{d}{du}\left(2u^2 - 3u + 8\right)$

$\boxed{2 \text{ pts}}$ (b) $f'(x)$, where $f(x) = x^{3.2} + 7x^{2.1} - 3x + 7$

$\boxed{2 \text{ pts}}$ (c) $f'(x)$, where $f(x) = x^{4/5} + 4x^{3/5} - 2x^{-1/5} + 5$

Section 3.4

4 pts 1. Let $f(x) = (4x^2 + 7)(2x^2 - 9)$. Find the value(s) of x for which the line tangent to the graph of f at $(x, f(x))$ is horizontal.

1 (bonus) 2. Find the first derivative of $f(x) = 3e^{2x}$.